

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(ELECTRICAL AND ELECTRONIC ENGINEERING).

EEE 203: CIRCUIT ANALYSIS I

DATE: Thursday, 10th December, 2015

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

Answer question ONE and TWO other questions.

QUESTION ONE

- a) Describe
- i) Ohms Law (3 marks)
 - ii) Kirchhoff's Current and Voltage laws (4 marks)
 - iii) The concept of EMF and potential difference (4 marks)
- b) While applying Kirchhoff's second law to closed circuit, it is important to assign proper sign to EMFs and voltage drops in the closed circuit.
- i) When is an Emf labeled positive? (2 marks)
 - ii) When is an IR drop across a resistor labeled negative? (2marks)
- c) A coil of resistance $100\ \Omega$ and inductance 0.15H is connected to a 250V , 50HZ Supply.
Calculate the capacitance to be connected in parallel with the coil so that overall p.f is unity (4marks)
- d) Calculate the value of the unknown resistance in Fig 1(d). The voltage drop across the first 500Ω is 2.5V . (5 marks)

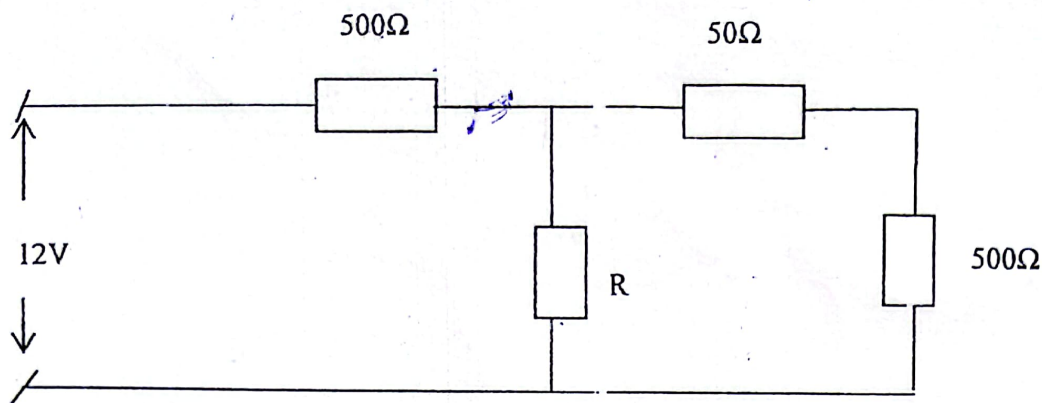


Fig 1 (d)

- e) Determine the value of the resistance R that is needed to have the current I to lag behind the applied voltage V by 90° in the circuit shown (Fig 1(e), where

$$Z_1 = 30 + j100\Omega$$

$$Z_R = 50 + j150\Omega$$

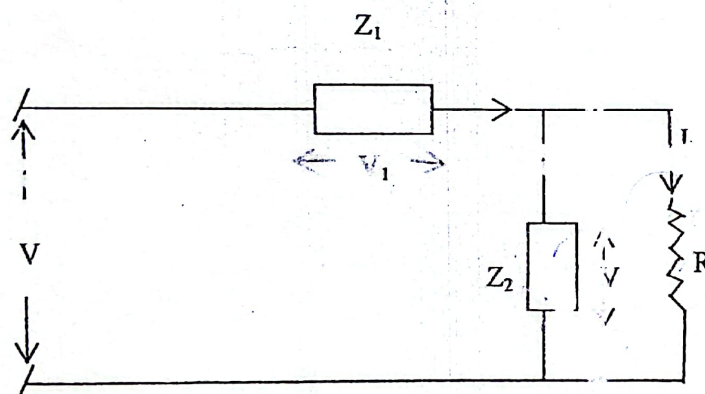


Fig 1 (e)

Handwritten notes:

$$\gamma_R = \frac{G}{\frac{R}{Q^2} + j0}$$

$$\frac{G}{Y} = 1 \quad \frac{Y}{Y} = 1$$

(6 marks)



QUESTION TWO

- a) Explain the terminologies in dc networks

- | | |
|---------------------|-----------|
| i) Active element | (1 mark) |
| ii) Passive element | (2 marks) |
| iii) Node | (1 mark) |
| iv) Junction | (1 mark) |
| v) Loop | (1 mark) |

EEE 203 10th DEC 2015

b) Define the following

i) Electrical Power

(1 mark)

ii) Electric Energy

(1 mark)

iii) Conductance

(1 mark)

c) State

i) Thevenin's theorem

(3 marks)

ii) Superposition theorem

(3 marks)

d) In the network shown Fig 2(c) find the different branch currents by superposition theorem.

(5 marks)

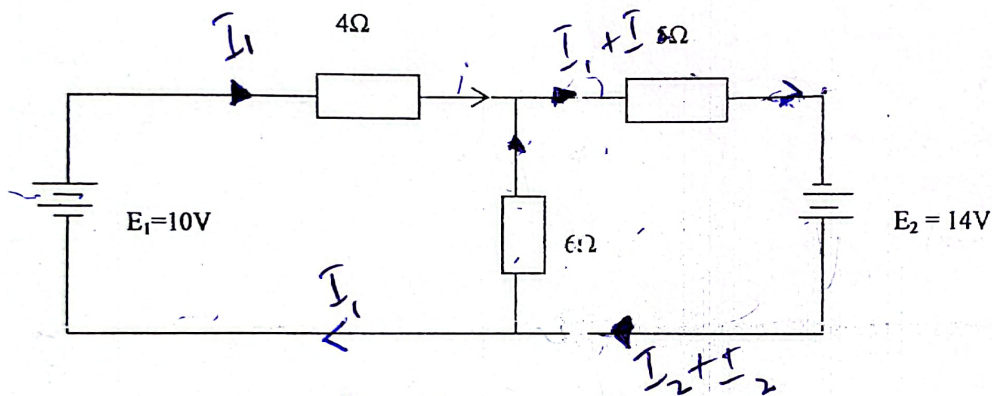


Fig 2 (c)

QUESTION THREE

a) Define these terms used in circuit analysis

i) Resistance (R)

(2 marks)

ii) Capacitance (C)

(2 marks)

iii) Inductance (L)

(2 marks)

Show how they relate to each other

(3 marks)

b) A coil of p.f 0.8 is connected in series with a $110\mu\text{F}$ Capacitor. The frequency is 50Hz. The p.d across the coil is found to be equal to the p.d across the capacitor, Calculate the resistance and inductance of the coil

(5 marks)

c) Using Norton's theorem, find the current flowing in the branch AB containing 6Ω of the network in the Figure 4 (c)

(6 marks)

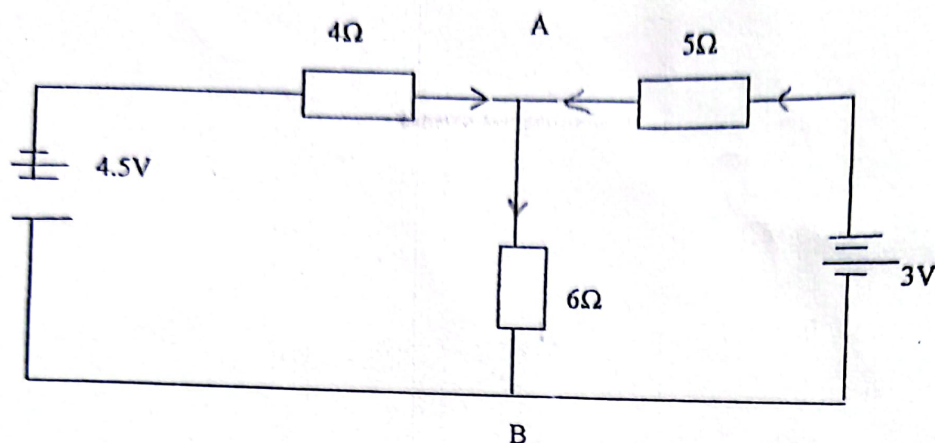


Fig 3 (c)

QUESTION FOUR

e) Explain

- i) The three main methods of solving A.C Parallel circuits (3 marks)
- ii) The converse of Thevenin's theorem (3 marks)
- iii) When is it appropriate to use Star/Delta Transformation (2 marks)

a) A voltage of $230 \angle 30^\circ$ V is applied to two circuits connected in parallel. The current in the branches are $20 \angle 60^\circ$ A and $40 \angle -30^\circ$ A.

Find

- i) The total impedance of the circuit (3 marks)
 - ii) Active and apparent power of each branch (2 marks)
- b) Two impedances $Z_1 = 2 + j3$ and $Z_2 = 3 - j4$ are connected in parallel across 250V, 50Hz supply.

Calculate

- i) The current in each branch (2 marks)
- ii) The line current (2 marks)
- iii) Circuit p.f (1 mark)
- iv) Power taken by the circuit (2 marks)

EEE 203 10th DEC 2015

QUESTION FIVE

a) Define the following terminologies

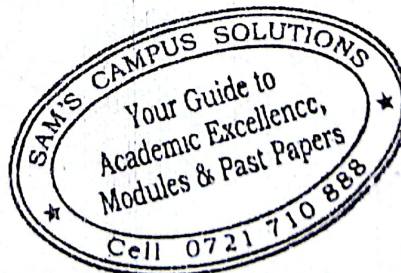
- i) Impedance (2 marks)
- ii) Admittance (1 mark)
- iii) Conductance (1 mark)
- iv) Susceptance (1 mark)

Show how they relate to each other and indicate their units (2 marks)

b) A 230V, 50Hz a.c supply is applied to a coil of 0.06H Inductance and 2.5Ω resistance connected in series with a $6.8\mu\text{F}$ capacitor.

Calculate

- i) Impedance (2 marks)
 - ii) Current (2 marks)
 - iii) Phase angle between current and voltage (1 mark)
 - iv) Power factor (1 mark)
 - v) Power consumed (1 mark)
- c) A single phase motor operating off a 400V, 50Hz supply is developing 10kw with an efficiency of 84% and Power factor of 0.7 lagging
- d) Calculate
- e) A) The input apparent power (2 marks)
 - f) The active and reactive component of current (2 marks)
 - g) The reactive power (in kilovars) (2 marks)

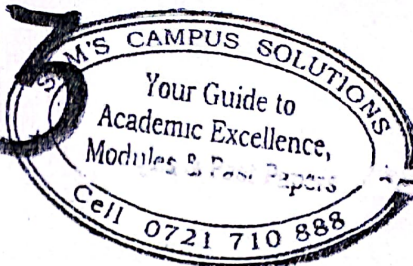




EEE



203



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(AGRICULTURAL AND BIOSYSTEMS ENGINEERING)

EEE 203: CIRCUIT THEORY

DATE: MONDAY, 18TH MARCH 2019

TIME: 11.00AM - 1.00PM

INSTRUCTIONS:

Answer Question ONE and any other TWO questions

Question ONE is compulsory and carries 30 marks

The remaining questions carry 20 marks each

QUESTION ONE

- (a) Define the following terms, show their units and how they relate to each other
- (i) Electric current (A) (2marks)
 - (ii) Potential difference (V) (2marks)
 - (iii) Specific Resistance or Resistivity (2marks)
 - (iv) Conductance (2marks)
- (b) Discuss the following
- (i) Factors upon which Resistance depends (4marks)
 - (ii) Effects of temperature on Resistance (3marks)
 - (iii) Concepts of EMF and Potential difference (4marks)
- (c) Four resistors are in parallel. The currents in the first three resistors are 4mA, 5mA and 6mA respectively. The voltage across the fourth resistor is 200 volts. The total power dissipated is 5watts. Determine the values of resistances of the branches and the total resistance. (6marks)
- (d) When a resistor is placed across a 230v supply, the current is 12A, What is the value of the resistor that must be placed in parallel to increase the load to 16A (5marks)

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Page 1 of 4

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QUESTION TWO

- (a) (i) State Kirchhoff's laws as applied to DC circuits (3marks)
 (ii) Discuss signs of Emfs and Voltage drops in a closed circuit (3marks)
 (iii) Method use to solving circuits by Kirchhoff's laws (4marks)
- (b) Calculate currents in the 20Ω resistor in fig 2(b)

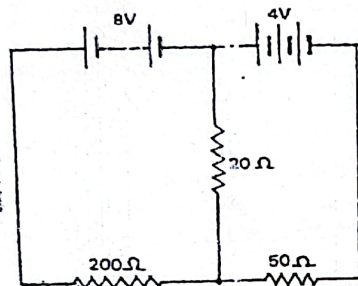
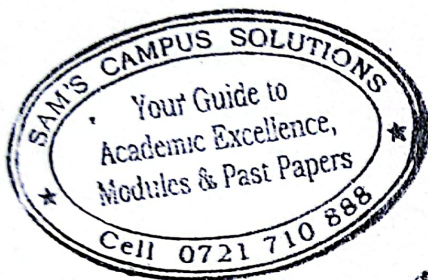


Fig 2(b)

(5marks)

- (c) Determine the currents I , I_1 , I_2 , I_3 flowing in the circuit Fig 2(c)

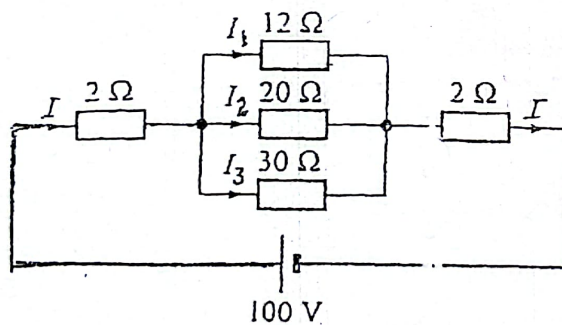


Fig 2(c)

(5marks)

QUESTION THREE

- (a) State the following theorems
- (i) Nortons theorem (3marks)
 - (ii) Superposition theorem (4marks)
 - (iii) Maxwells mesh current method (3marks)
- (b) (i) Use the superposition theorem to find current through R_1 in the circuit shown Fig 3(b) (5marks)
- (ii) Use Thevenins theorem to find the current through R_1 in the same circuit Fig 3(b) (5marks)

EEE 203

18/03/2019

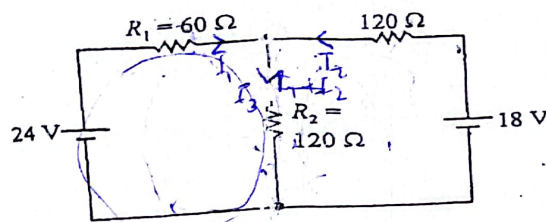
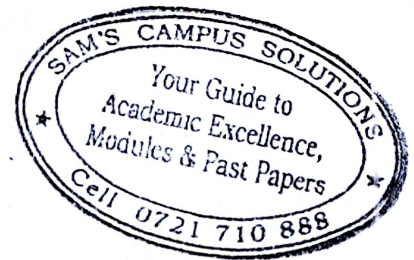


Fig 3(b)

QUESTION FOUR

(a) Explain the following

- (i) An ideal voltage source and ideal current source (3marks)
- (ii) Nortons theorem being the converse of Thevenin's theorem (3marks)
- (iii) When it is appropriate to use star delta /delta star transformation (2marks)
- (iv) The significance of power factor (3marks)
- (v) The significance of operator j (3marks)

(b) Determine the load current i_{in} in the branch CD and EF in the circuit below Fig 4(b) by converting first the star into Delta. All resistance are in ohms (6marks)

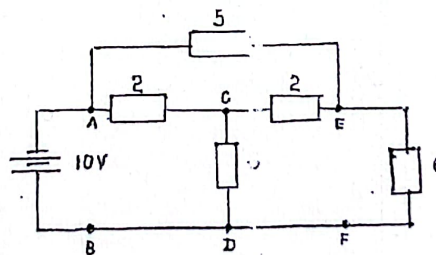


Fig. 4(b)



QUESTION FIVE

- (a) A coil of p.f 0.8 is connected in series with a $110\mu F$ capacitor. The supply frequency is 50 Hz. The p.d across the coil is found to be equal to the p.d across the capacitor. Calculate the resistance and inductance of the coil (4marks)

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18/03/2019

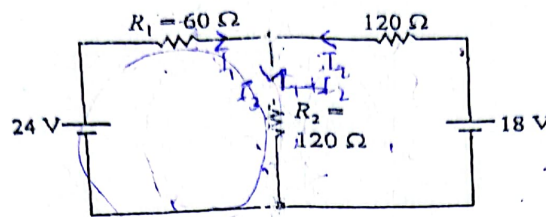
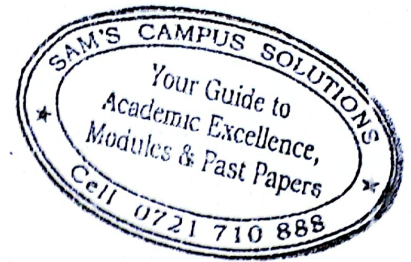


Fig 3(b)

QUESTION FOUR

(a) Explain the following

- (i) An ideal voltage source and ideal current source (3marks)
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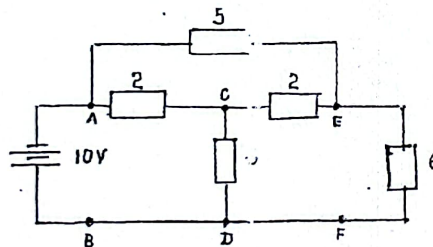


Fig. 4(b)



QUESTION FIVE

- (a) A coil of p.f 0.8 is connected in series with a $110\mu\text{F}$ capacitor. The supply frequency is 50 Hz. The p.d across the coil is found to be equal to the p.d across the capacitor. Calculate the resistance and inductance of the coil (4marks)

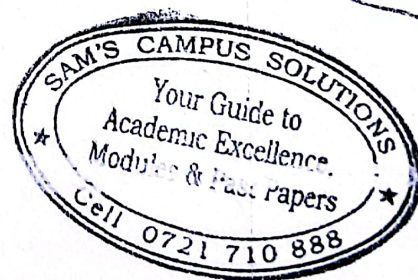
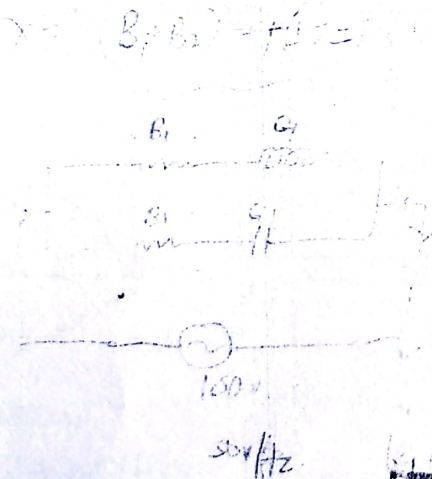
- (b) A coil of resistance 10Ω and inductance $0.1H$ is connected in series with a capacitor $150mF$ across a $200V$, $50Hz$ supply.

Calculate:

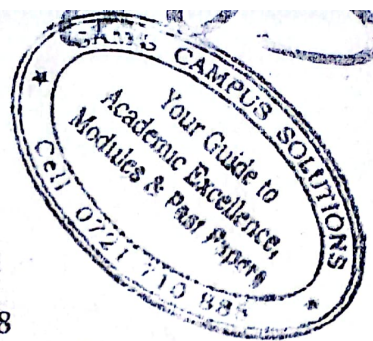
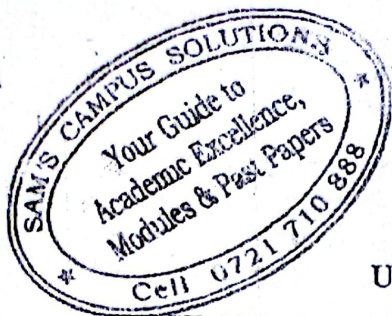
- (i) Impedance (2marks)
- (ii) Power factor (1mark)
- (iii) Current (1mark)
- (iv) Voltage across coil (1mark)
- (v) Voltage across capacitor (1mark)

- (c) Two impedances $Z_1 = (10+j5)\Omega$ and $Z_2 = (25-j10)\Omega$ are connected in parallel to a $100V$ $50Hz$ supply

- Find (i) Circuit admittance (2marks)
- (ii) Circuit current (2marks)
- (iii) The phase angle between circuit current and applied voltage (2marks)
- (iv) Circuit p.f (2marks)
- (v) Power taken by the circuit (2marks)



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UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE

EEE 203: CIRCUIT ANALYSIS

DATE: Monday, 26th February 2018

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

Answer Question ONE and any other TWO Questions

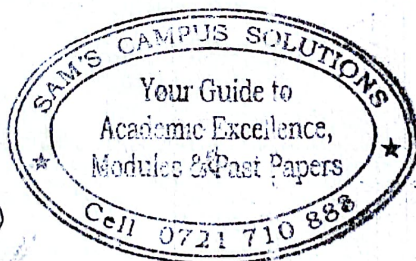
Question ONE is Compulsory and carries 30 Marks

The remaining questions carry 20 Marks)

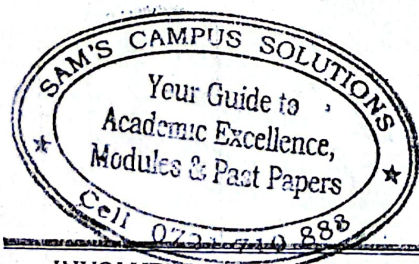
QUESTION ONE

a) Define the following terms, show their units and how they relate to each other.

- | | |
|------------------------|-----------|
| i) Capacitance (C) | (3 Marks) |
| ii) Inductance (L) | (3 Marks) |
| iii) Impedance (Z) | (3 Marks) |
| iv) Power factor (p.f) | (3 Marks) |



- b) i) State Kirchoff's laws as applied to dc circuits. (4 Marks)
- ii) applying Kirchhoff's voltage law to a network path
- When is an emf labelled positive (2 Marks)
 - IR drop across a resistor labelled positive (2 Marks)



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- c) By using Kirchoff's laws, find the currents in various resistors in the circuit shown Fig 1(c)

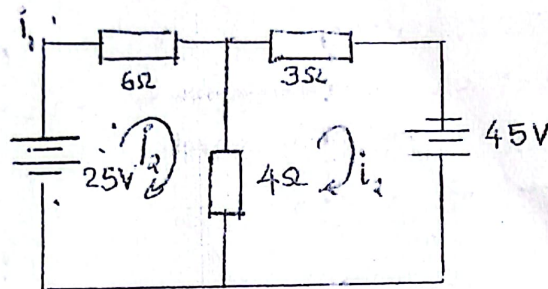


Fig 1(c)

(6 Marks)

- d) Calculate the value of the unknown resistor R in Fig. 1(d). The voltage drop across the 500Ω resistor is 2.5V .

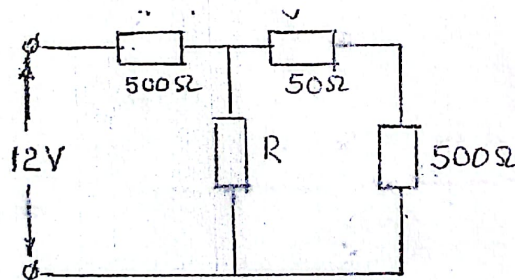


Fig 1(d)

(4 Marks)

QUESTION TWO

- a) Define and then identify the following in the diagram Fig. (2a)

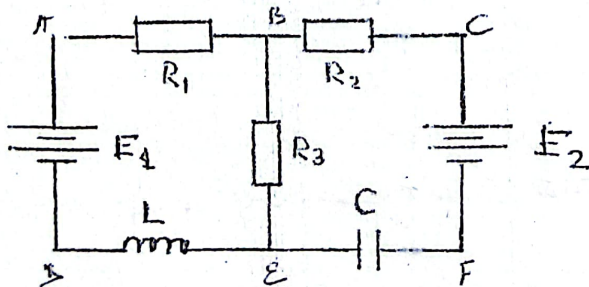
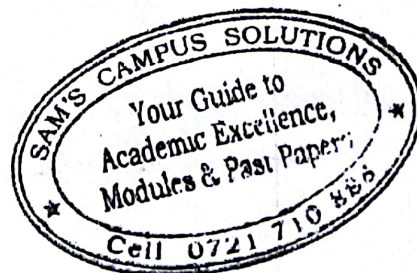
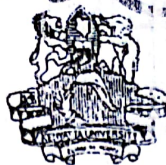


Fig. 2(a)





KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2016/2017
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR
OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)
EEE 203: CIRCUIT THEORY

DATE: TUESDAY 13TH DECEMBER 2016 **TIME:** 2.00 P.M. - 4.00 P.M.

INSTRUCTIONS

Answer question 1 and any other 2

- a) Explain the following terminologies used in current electricity.
- i) Electric power (2 marks)
 - ii) Electrical Energy (2 marks)
 - iii) Concept of EMF and potential difference (3 marks)
- b) A battery has an emf of 12.8 Volts and supplies a current of 3.24 A.
- Calculate i) the resistance of the circuit (2 marks)
 - ii) How many coulombs leave the battery in 5 minutes. (3 marks)
- c) Analyze the special case of two resistors in a parallel circuit
- i) Total resistance (R_T) (2 marks)
 - ii) Branch currents (2 marks)
- d) A 50-Ohm resistor is in parallel with a 100-ohm resistor. The current in 50 Ω resistor is 7.2 A. What is the value of third resistance to be added in parallel to make the line current 12.1 A. (4 marks)
- e) When a resistor is placed across a 230V supply the current is 12A. What is the value of a resistor that must be placed in parallel to increase the load current to 16A. (4 marks)

- f) Using Kirchhoff's laws find the currents through various resistors in the circuit. (6 marks)

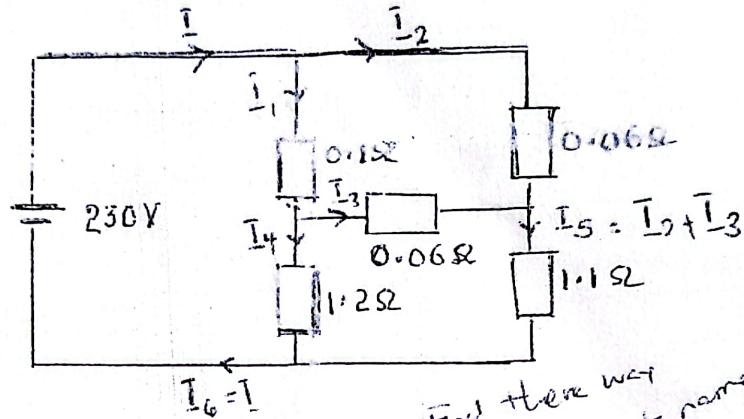


Fig 1(f)

Question Two

- a) Explain the following terms in circuit analysis,

- Superposition theorem. (2 marks)
- Kirchhoff's laws (3 marks)
- Signs of emf's and voltage drops. (3 marks)

- b) Using Kirchhoff's laws.

Find the current in various resistors in the circuit shown.

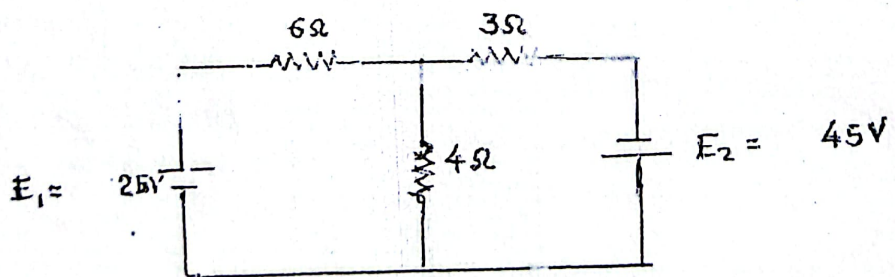


Fig 2(b)

(6 marks)

- c) Using superposition theorem find the currents in the different branches of the network.
All resistances are in Ohms.

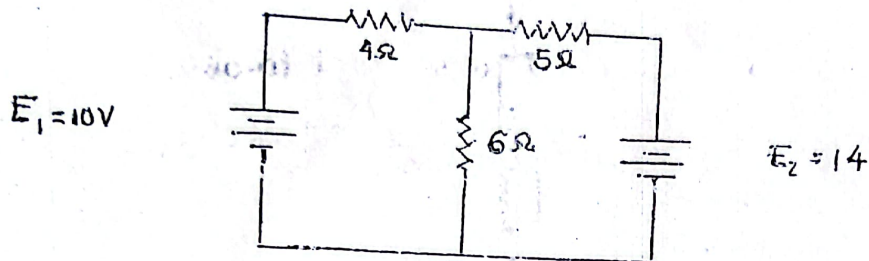


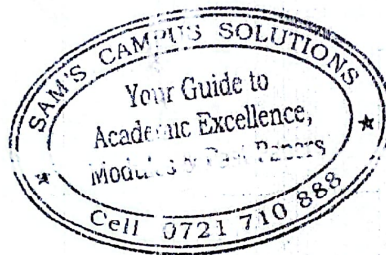
Fig 3(c)

(6 marks)

Question Three

- a) State the following

- Thevenin's theorem.
- Norton's theorem.



(3 marks)

(3 marks)

- b) Using Norton's theorem, find the current in the branch AB containing 6Ω resistor of the network shown in Fig 3(b)

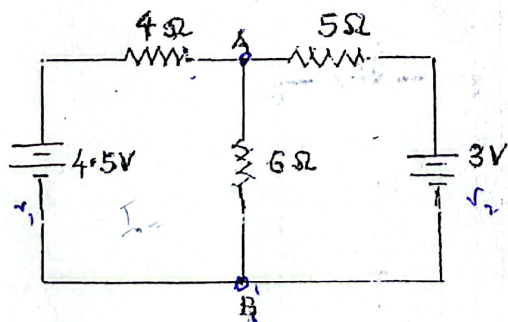


Fig 3(b)

(6 marks)

- c) A resistance of 6Ω and an inductance of $0.03H$ is connected across a $50V$, $60Hz$ supply.

Calculate

- i) The current (2 marks)
 ii) The phase angle between current and applied voltage. (2 marks)
 iii) The apparent power. (2 marks)
 iv) Active power. (2 marks)

Question Four

- a) Explain the following a.c. terminologies
- i) Cycle (1 mark)
 ii) Frequency (1 mark)
 iii) Amplitude (2 marks)
 iv) RMS value of sinusoidal current (2 marks)
- b) i) Explain the difference between Thevenin's and Norton's theorems. (3 marks)
 ii) The reasons why sinusoidal wave rather than square or triangle wave is chosen for Generation and Transmission of power (3 marks)
- c) Determine the load current in the branch EF in the circuit shown. Fig(c). Use star/delta transformation

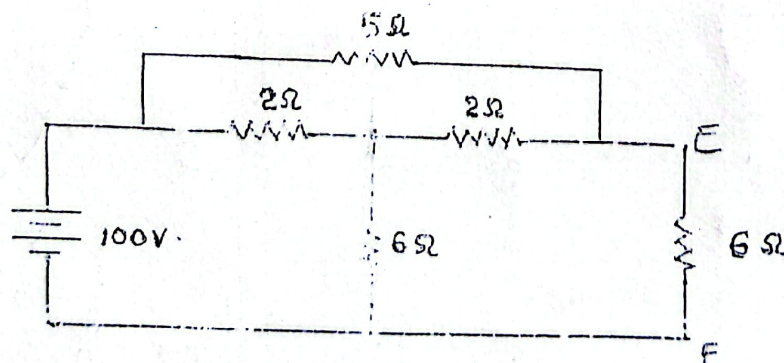
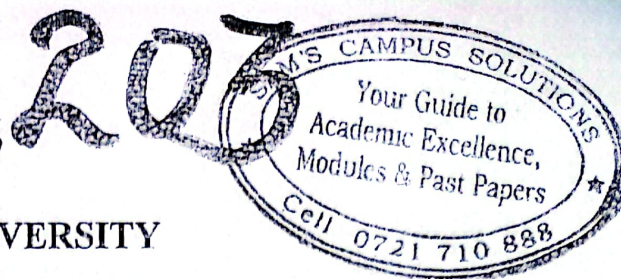
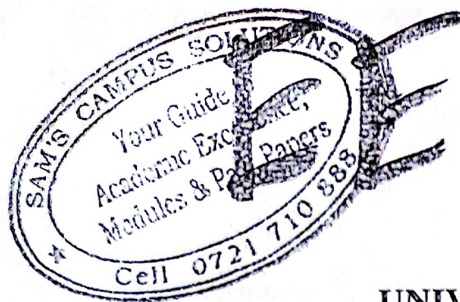


Fig 4(c)

(8 marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

**EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(AGRICULTURAL AND BIOSYSTEMS ENGINEERING)**

EEE 203: CIRCUIT THEORY

DATE: MONDAY, 18TH MARCH 2019

TIME: 11.00AM - 1.00PM

INSTRUCTIONS:

Answer Question ONE and any other TWO questions

Question ONE is compulsory and carries 30 marks

The remaining questions carry 20 marks each

QUESTION ONE

(a) Define the following terms, show their units and how they relate to each other

- (i) Electric current (A) (2marks)
- (ii) Potential difference (V) (2marks)
- (iii) Specific Resistance or Resistivity (2marks)
- (iv) Conductance (2marks)

(b) Discuss the following

- (i) Factors upon which Resistance depends (4marks)
- (ii) Effects of temperature on Resistance (3marks)
- (iii) Concepts of EMF and Potential difference (4marks)

(c) Four resistors are in parallel. The currents in the first three resistors are 4mA, 5mA and 6mA respectively. The voltage across the fourth resistor is 200 volts. The total power dissipated is 5watts. Determine the values of resistances of the branches and the total resistance. (6marks)

(d) When a resistor is placed across a 230v supply, the current is 12A, What is the value of the resistor that must be placed in parallel to increase the load to 16A (5marks)

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QUESTION TWO

- (a) (i) State Kirchhoff's laws as applied to DC circuits (3marks)
 (ii) Discuss signs of Emfs and Voltage drops in a closed circuit (3marks)
 (iii) Method use to solving circuits by Kirchhoff's laws (4marks)
- (b) Calculate currents in the 20Ω resistor in fig 2(b)

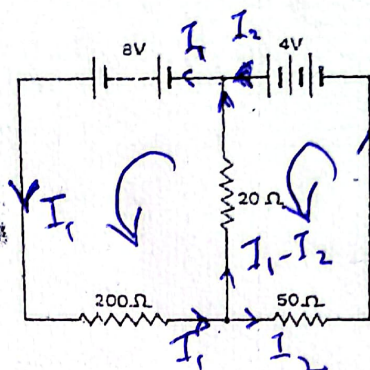
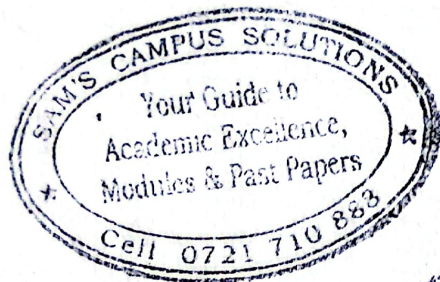


Fig 2(b)

$$\begin{aligned} 8 + -200I_1 - 20(I_1 - I_2) &= 0 \\ 4 + 20(I_1 - I_2) - 50I_2 &= 0 \\ 8 - 220I_1 + 20I_2 &= 0 \\ (200I_1 - 20I_2 &= 8) \\ - (20I_1 - 70I_2 &= -4) \end{aligned}$$

- (c) Determine the currents I , I_1 , I_2 , I_3 flowing in the circuit Fig 2(c)

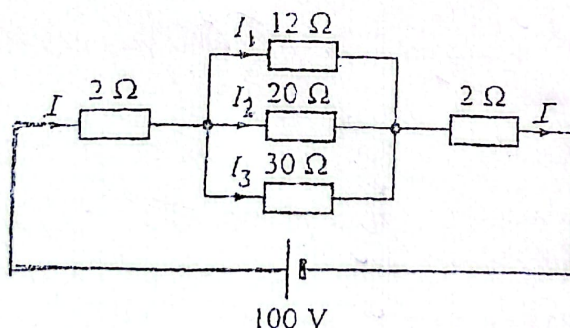
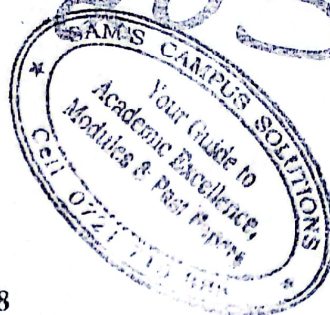


Fig 2(c)

$$\begin{aligned} -20I_2 &= 8 \\ -700I_2 &= -40 \end{aligned}$$

QUESTION THREE

- (a) State the following theorems (3marks)
 (i) Nortons theorem (3marks)
 (ii) Superposition theorem (4marks)
 (iii) Maxwells mesh current method (3marks)
- (b) (i) Use the superposition theorem to find current through R_1 in the circuit shown Fig 3(b) (5marks)
 (ii) Use Thevenins theorem to find the current through R_1 in the same circuit Fig 3(b) (5marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE

EEE 203: CIRCUIT ANALYSIS

DATE: Monday, 26th February 2018

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

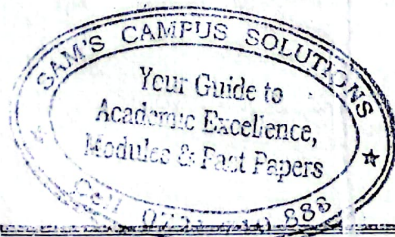
Answer Question ONE and any other TWO Questions

Question ONE is Compulsory and carries 30 Marks

The remaining questions carry 20 Marks)

QUESTION ONE

- a) Define the following terms, show their units and how they relate to each other.
- | | |
|------------------------|-----------|
| i) Capacitance (C) | (3 Marks) |
| ii) Inductance (L) | (3 Marks) |
| iii) Impedance (Z) | (3 Marks) |
| iv) Power factor (p.f) | (3 Marks) |
- b) i) State Kirchhoff's laws as applied to dc circuits. (4 Marks)
- ii) applying Kirchhoff's voltage law to a network path
- When is an emf labelled positive (2 Marks)
 - IR drop across a resistor labelled positive (2 Marks)



INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

- c) By using Kirchoff's laws, find the currents in various resistors in the circuit shown Fig 1(c)

1(c)

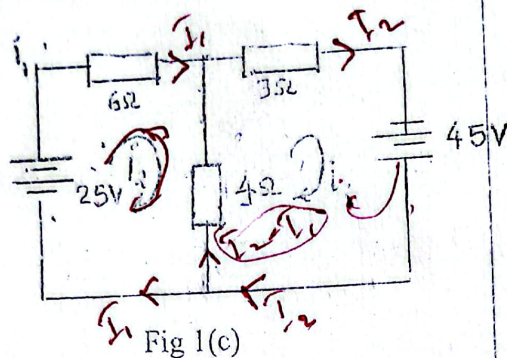


Fig 1(c)

(6 Marks)

- d) Calculate the value of the unknown resistor R in Fig. 1(d) The voltage drop across the 500Ω resistor is 2.5v.

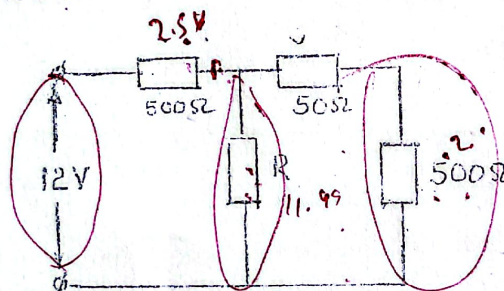


Fig 1(d)

(4 Marks)

QUESTION TWO

2a)

Define and then identify the following in the diagram Fig. (2a)

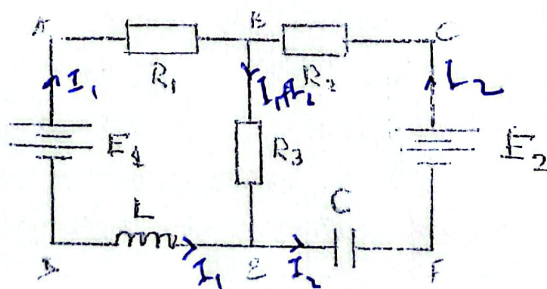
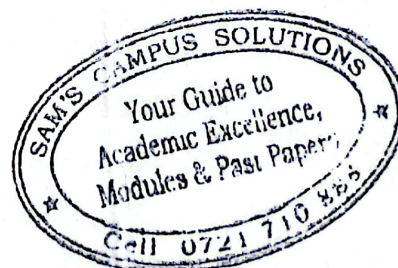


Fig 2(a)



- (b) A coil of resistance 10Ω and inductance $0.1H$ is connected in series with a capacitor $150mF$ across a $200V$, $50Hz$ supply.

Calculate:

- (i) Impedance (2marks)
- (ii) Power factor (1mark)
- (iii) Current (1mark)
- (iv) Voltage across coil (1mark)
- (v) Voltage across capacitor (1mark)

- (c) Two impedances $Z_1 = (10+j5)\Omega$ and $Z_2 = (25-j10)\Omega$ are connected in parallel to a $100V$ $50Hz$ supply

- Find (i) Circuit admittance (2marks)
- (ii) Circuit current (2marks)
- (iii) The phase angle between circuit current and applied voltage (2marks)
- (iv) Circuit p.f (2marks)
- (v) Power taken by the circuit (2marks)

$$Y_1 = \frac{1}{Z_1} \quad Z_1 = R_1 + jX_L$$

$$Z_1 = (R + jX_L)$$

$$Y_1 = \frac{R_1}{R_1^2 + X_L^2} + \frac{jX_L}{R_1^2 + X_L^2}$$



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EEF 203

18/03/2019

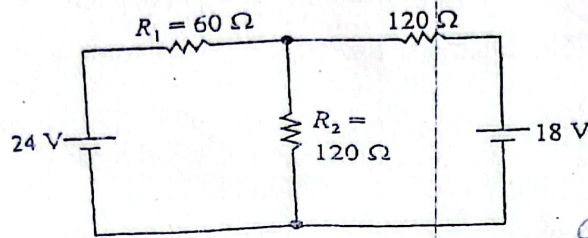


Fig 3(b)

QUESTION FOUR

- (a) Explain the following
- (i) An ideal voltage source and ideal current source (3marks)
 - (ii) Norton's theorem being the converse of Thevenin's theorem (3marks)
 - (iii) When it is appropriate to use star delta /delta star transformation (2marks)
 - (iv) The significance of power factor (3marks)
 - (v) The significance of operator j (3marks)
- (b) Determine the load current i_{in} in the branch CD and EF in the circuit below Fig 4(b) by converting first the star into Delta. All resistance are in ohms (6marks)

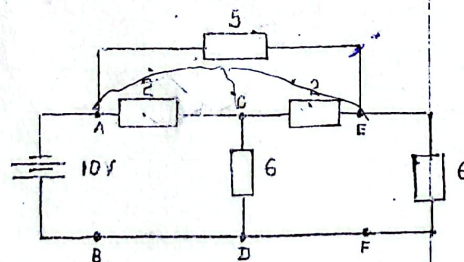
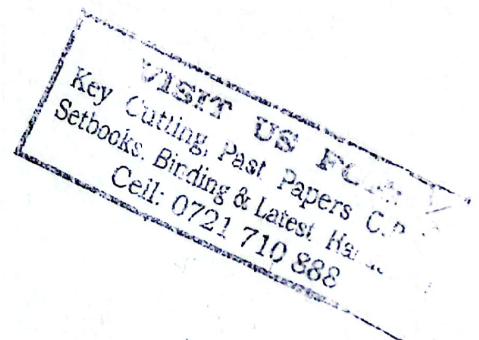


Fig. 4(b)



QUESTION FIVE

- A coil of p.f 0.8 is connected in series with a $110\mu F$ capacitor. The supply frequency is 50 Hz. The p.d across the coil is found to be equal to the p.d across the capacitor. Calculate the resistance and inductance of the coil (4marks)

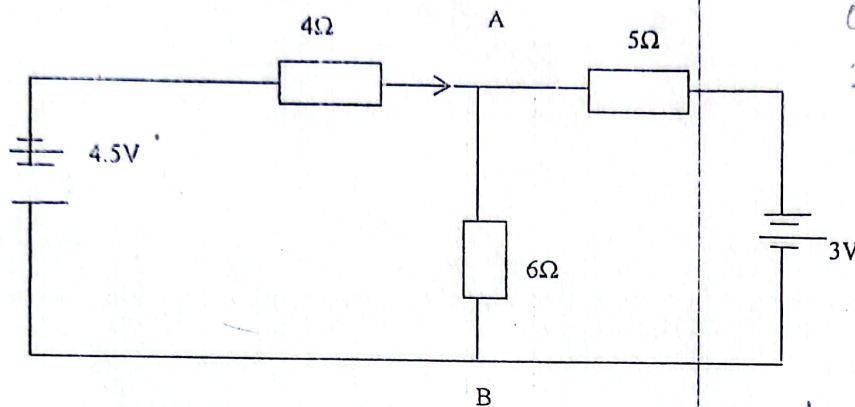


Fig 3 (c)

0.2445

$Z = R + jX_L$
 $Y = G - jB$
 $G = \frac{R}{Z^2}$
 $B = \frac{X_L}{Z^2}$
 $Y = \frac{1}{Z}$

QUESTION FOUR

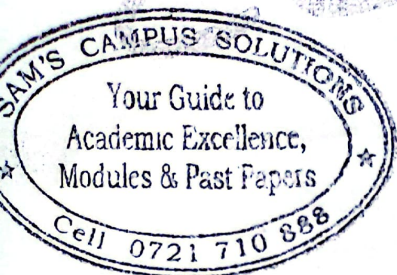
- a) Explain
 - i) The three main methods of solving A.C Parallel circuits (3 marks)
 - ii) The converse of Thevenin's theorem (3 marks)
 - iii) When is it appropriate to use Star/Delta Transformation (2 marks)
- a) A voltage of $230 \angle 30^\circ$ V is applied to two circuits connected in parallel. The current in the branches are $20 \angle 60^\circ$ A and $40 \angle -30^\circ$ A.
Find
 - i) The total impedance of the circuit (3 marks)
 - ii) Active and apparent power of each branch (2 marks)
- b) Two impedances $Z_1 = 2 + j3$ and $Z_2 = 3 - j4$ are connected in parallel across 250V, 50Hz supply.
Calculate
 - i) The current in each branch (2 marks)
 - ii) The line current (2 marks)
 - iii) Circuit p.f (1 mark)
 - iv) Power taken by the circuit (2 marks)

$$I_1 = \frac{V}{Z_1} = \frac{250 \angle 0^\circ}{2 + j3} = \frac{250}{13} \angle -56.3^\circ$$

$$I_2 = \frac{V}{Z_2} = \frac{250 \angle 0^\circ}{3 - j4} = \frac{250}{5} \angle 53.1^\circ$$

$$I_L = I_1 + I_2 = \frac{250}{13} \angle -56.3^\circ + \frac{250}{5} \angle 53.1^\circ$$

$$P.F. = \cos \phi = \cos 12.6^\circ = 0.975$$



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(ELECTRICAL AND ELECTRONIC ENGINEERING)

EEE 203: CIRCUIT ANALYSIS I

DATE: Thursday, 10th December, 2015

TIME: 11.00 a.m. - 1.00 p.m.

$$V = IR$$
$$V_{ma} = I_{ma} R$$
$$V = \frac{I}{1/2} R$$

INSTRUCTIONS:

Answer question ONE and TWO other questions.

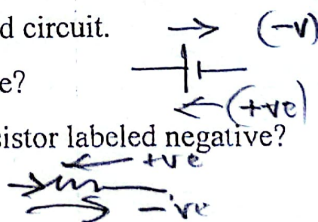
QUESTION ONE

a) Describe

- i) Ohms Law (3 marks)
- ii) Kirchhoff's Current and voltage laws (4 marks)
- iii) The concept of EMF and potential difference (4 marks)

b) While applying Kirchhoff's second law to closed circuit, it is important to assign proper sign to EMFs and voltage drops in the closed circuit.

- i) When is an Emf labeled positive? (2 marks)
- ii) When is an IR drop across a resistor labeled negative? (2marks)



~~10~~ A coil of resistance 100Ω and inductance $0.15H$ is connected to a $250V$, $50HZ$ Supply.

Calculate the capacitance to be connected in parallel with the coil so that overall p.f is unity

$$Z_1^2 = R^2 + X_L^2 = 100 + j0.15 \quad (4marks)$$

$$Z_L = R_L$$

- d) Calculate the value of the unknown resistance in Fig 1(d). The voltage drop across the first 500Ω is $2.5V$. (5 marks)

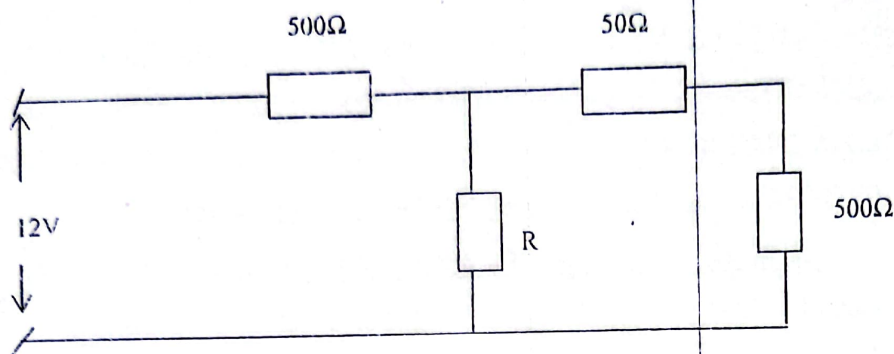


Fig 1 (d)

9) Determine the value of the resistance R that is needed to have the current I to lag behind the applied voltage V by 90° in the circuit shown (Fig 1(e), where

$$Z_1 = 30 + j100\Omega$$

$$Z_R = 50 + j150\Omega$$

(6 marks)

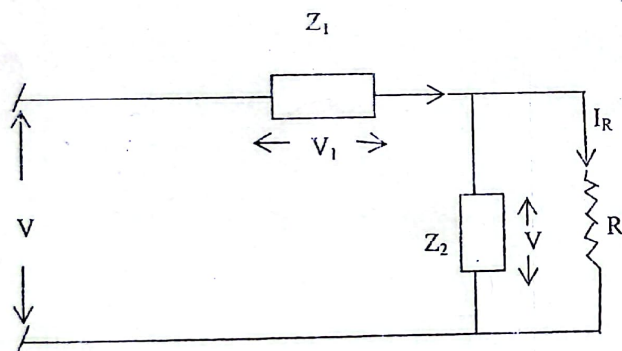
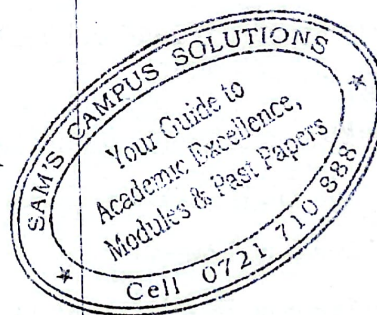


Fig 1 (e)



QUESTION TWO

a) Explain the terminologies in dc networks

i) Active element

(1 mark)

ii) Passive element

(2 marks)

iii) Node

(1 mark)

iv) Junction

(1 mark)

v) Loop

(1 mark)

QUESTION FIVE

a) Define the following terminologies

- i) Impedance (2 marks)
- ii) Admittance (1 mark)
- iii) Conductance (1 mark)
- iv) Susceptance (1 mark)

Show how they relate to each other and indicate their units

(2 marks)

b) A 230V, 50Hz a.c supply is applied to a coil of 0.06H Inductance and 2.5Ω resistance connected in series with a $6.8\mu\text{F}$ capacitor.

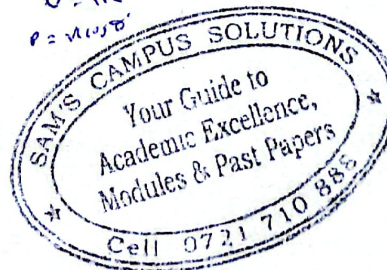
Calculate

- i) Impedance (2 marks)
 - ii) Current (2 marks)
 - iii) Phase angle between current and voltage (1 mark)
 - iv) Power factor (1 mark)
 - v) Power consumed (1 mark)
- c) A single phase motor operating off a 400V, 50Hz supply is developing 10kw with an efficiency of 84% and Power factor of 0.7 lagging
- d) Calculate (2 marks)
- e) A) The input apparent power (2 marks)
- f) The active and reactive component of current (2 marks)
- g) The reactive power (in kilovars) (2 marks)

$$S = VI^* = \frac{V_{LMS}}{CSO}$$

$$P = VI \cos \phi$$

$$Q = VI \sin \phi$$



- e) Using superposition theorem find the currents in the different branches of the network.
All resistances are in Ohms.

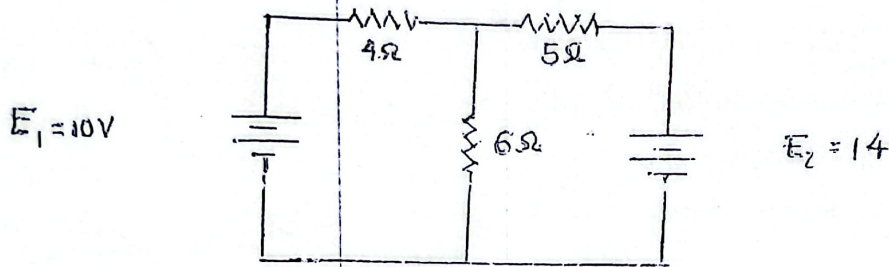


Fig 3(c)

(6 marks)

Question Three

- a) State the following

- i) Thevenins theorem.
- ii) Norton's theorem.



(3 marks)

(3 marks)

- b) Using Nortons theorem, find the current in the branch AB containing 6 Ω resistor of the network shown in Fig 3(b)

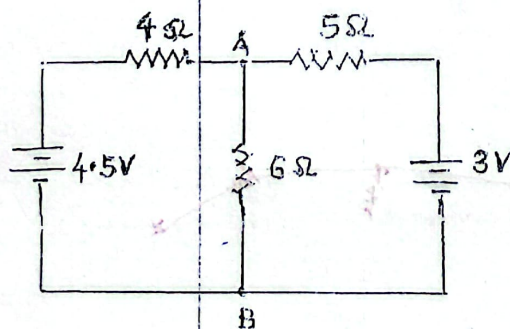


Fig 3(b)

(6 marks)

- c) A resistance of 6Ω and an inductance of $0.03H$ is connected across a $50V$, $60Hz$ supply.

Calculate

- The current (2 marks)
- The phase angle between current and applied voltage. (2 marks)
- The apparent power. (2 marks)
- Active power. (2 marks)

Question Four

Explain the following a.c. terminologies

- Cycle (1 mark)
 - Frequency (1 mark)
 - Amplitude (2 marks)
 - RMS value of sinusoidal current (2 marks)
- b) i) Explain the difference between Thevenin's and Norton's theorems. (3 marks)
- ii) The reasons why sinusoidal wave rather than square or triangle wave is chosen for Generation and Transmission of power (3 marks)
- c) Determine the load current in the branch EF in the circuit shown. Fig(c). Use star/delta transformation

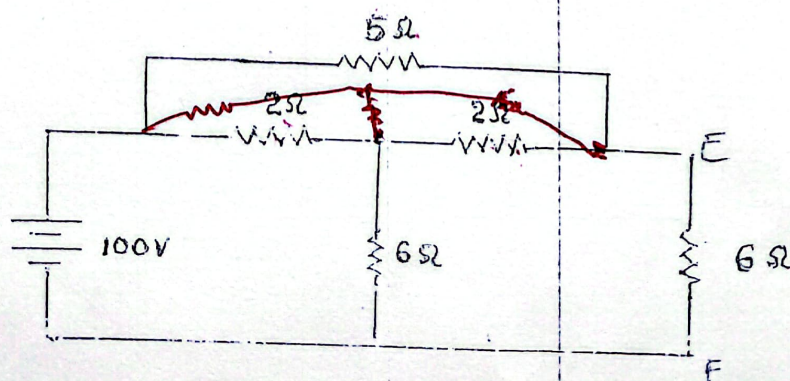
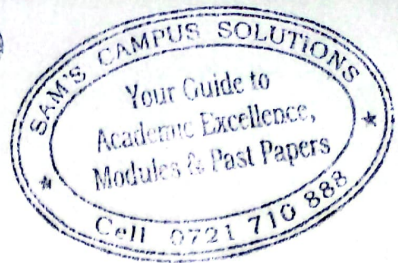


Fig 4(c)

(8 marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2016/2017
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR
OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING)
EEE 203: CIRCUIT THEORY

DATE: TUESDAY 13TH DECEMBER 2016

TIME: 2.00 P.M. - 4.00 P.M.

INSTRUCTIONS

Answer question 1 and any other 2

- a) Explain the following terminologies used in current electricity.
- i) Electric power (2 marks)
 - ii) Electrical Energy (2 marks)
 - iii) Concept of EMF and potential difference (3 marks)
- b) A battery has an emf of 12.8 Volts and supplies a current of 3.24 A. Calculate
- i) the resistance of the circuit (2 marks)
 - ii) How many coulombs leave the battery in 5 minutes. (3 marks)
- c) Analyze the special case of two resistors in a parallel circuit
- i) Total resistance (R_T) (2 marks)
 - ii) Branch currents (2 marks)
- d) A 50-Ohm resistor is in parallel with a 100-ohm resistor. The current in 50 Ω resistor is 7.2 A. What is the value of third resistance to be added in parallel to make the line current 12.1A. (4 marks)
- e) When a resistor is placed across a 230V supply the current is 12A. What is the value of a resistor that must be placed in parallel to increase the load current to 16A. (4 marks)

- f) Using Kirchhoff's laws find the currents through various resistors in the circuit. (6 marks)

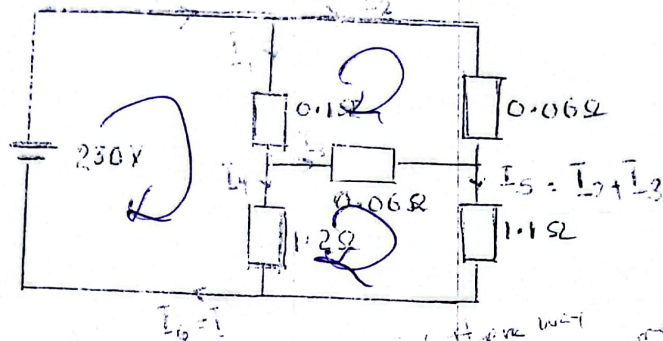


Fig 1(f)

*This time with
it's sound
at your great name*

Question Two

- a) Explain the following terms in circuit analysis,
- Superposition theorem (2 marks)
 - Kirchhoff's laws (3 marks)
 - Signs of emf's and voltage drops. (3 marks)
- b) Using Kirchhoff's laws.
Find the current in various resistors in the circuit shown.

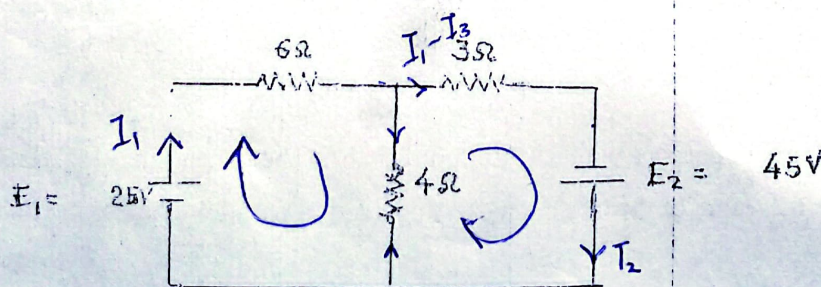


Fig 2(b)

(6 marks)